

Payload / method	Bytes / repeats	Reference encode / decode (s)	Graph encode / decode (s)	Pair speedup	Recovered B/s reference $\rightarrow$ graph	Equivalence
T1 / Fixed	32 / 3	88.97 / 89.28	17.38 / 17.28	$5.14\times$	$0.180 \rightarrow 0.923$	Exact
T1 / Gated	32 / 1	90.17 / 90.07	17.38 / 17.23	$5.21\times$	$0.178 \rightarrow 0.925$	Exact
T1 / A1	32 / 1	91.83 / 91.03	17.83 / 17.28	$5.21\times$	$0.175 \rightarrow 0.911$	Exact
T4 / Fixed	96 / 3	89.12 / 89.09	17.34 / 17.29	$5.15\times$	$0.539 \rightarrow 2.772$	Exact
T5 / Fixed	128 / 3	88.76 / 88.24	17.36 / 17.27	$5.11\times$	$0.723 \rightarrow 3.696$	Exact

Matched reference versus CUDA graph benchmark. Three development payloads, row1; timing repetitions are not new independent research observations. Fixed has three matched repetitions per payload; gated/A1 each have one T1 comparison. Latencies are means of full charged fresh processes, including loading, graph setup and teardown. Speedup is the ratio of matched mean charged pair times; per-row throughput is the accepted mean of individual exact-source-bytes/pair-second ratios (the Figure 4 pooled headline instead divides summed bytes by summed times). All 11 comparisons/22 recoveries match exact ID/order/probability stream hashes, PNG file bytes and recovered source bytes. Peak PyTorch allocated memory: $484.479 \rightarrow 473.849$ MiB; reserved: $516 \rightarrow 744$ MiB; sampled process maximum: $912 \rightarrow 1,176$ MiB. These are distinct, overlapping measures, not additive; per-combination measurements and nested CUDA-event/loading times are in the full-precision CSV. Tested RTX 5000 Ada, PyTorch 2.5.1+cu124 and unchanged checkpoint/float32 precision; no cross-hardware performance claim.